PGCP-AC August 26
Database Assignment No-3

1. Write a query to calculate the total salary of all employees.

```
mysql> select sum(salary) from employee;
+-------------+
| sum(salary) |
+-------------+
|      338500 |
+-------------+
1 row in set (0.11 sec)
```

2. Write a query to find the average salary of employees in each department using GROUP BY.

```
mysql> select dept_id, avg(salary)
    -> from employee
    -> group by dept_id;
+---------+-------------+
| dept_id | avg(salary) |
+---------+-------------+
|     101 |  74250.0000 |
|     102 |  57500.0000 |
|     103 |  40000.0000 |
|     105 |  35000.0000 |
+---------+-------------+
4 rows in set (0.00 sec)
```

3. Write a query to count the total number of employees in each department.

```
mysql> select dept_id, count(emp_id)
    -> from employee
    -> group by dept_id;
+---------+---------------+
| dept_id | count(emp_id) |
+---------+---------------+
|     101 |             2 |
|     102 |             2 |
|     103 |             1 |
|     105 |             1 |
+---------+---------------+
4 rows in set (0.02 sec)
```

4. Write a query to display departments having more than 2 employees using HAVING clause.

```
mysql> select dept_id from employee
    -> having count(*) > 2;
+---------+
| dept_id |
+---------+
|     101 |
+---------+
1 row in set (0.02 sec)
```

5. Write a query to list distinct department locations from the Dept table.

```
mysql> select distinct location from department;
+----------+
| location |
+----------+
| Mumbai   |
| Pune     |
| Dehli    |
+----------+
3 rows in set (0.00 sec)
```

6. Write a query to display the highest salary among all employees.

```
mysql> select max(salary) as highest_salary
    -> from employee;
+----------------+
| highest_salary |
+----------------+
|          77000 |
+----------------+
1 row in set (0.01 sec)
```

7. Write a query to display all total number of employee in table.

```
mysql> select count(emp_id) as total_employee
    -> from employee;
+----------------+
| total_employee |
+----------------+
|              6 |
+----------------+
1 row in set (0.10 sec)
```

8. Write a query to find employees whose name starts with 'A' using LIKE operator.

```
mysql> select * from  employee
    -> where emp_name like 'a%';
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
1 row in set (0.04 sec)
```

9. Write a query to find employees whose name ends with 'n' using LIKE operator.

```
mysql> select * from employee
    -> where emp_name like '%n';
Empty set (0.02 sec)
```

10. Write a query to find employees whose name contains 'ra' using LIKE operator.

```
mysql> select * from employee
    -> where emp_name like '%ra%';
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
1 row in set (0.00 sec)
```

11. Write a query to display all employees sorted by their Salary in descending order.

```
mysql> select * from employee
    -> order by salary desc;
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
6 rows in set (0.03 sec)
```

12. Write a query to display all employees sorted by DeptID ascending and then Salary descending.

```
mysql> select * from employee
    -> order by emp_id asc, salary desc;
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
6 rows in set (0.00 sec)
```

13. Write a query to find employees whose salary is between 30,000 and 60,000.

```
mysql> select * from employee
    -> where salary between 30000 and 60000;
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
4 rows in set (0.03 sec)
```

14. Write a query to display all employees whose DeptID is in (10, 20, 30).

```
mysql> select * from employee
    -> where dept_id in (101, 102, 103);
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |
+--------+----------+---------+--------+--------------+--------+
5 rows in set (0.03 sec)
```

15. Write a query to display Min salary of employee.

```
mysql> select min(salary) from employee;
+-------------+
| min(salary) |
+-------------+
|       35000 |
+-------------+
1 row in set (0.00 sec)
```

16. Write a query to display employees whose JoiningDate is between '2020-01-01' and '2021-12-31'.

```
mysql> select * from employee
    -> where joining_date between '2020-01-01' and '2021-12-31';
Empty set (0.02 sec)
```

17. Write a query to display employees whose Salary is NULL.

```
mysql> select * from employee
    -> where salary is null;
Empty set (0.00 sec)
```

18. Write a query to display employees whose Salary is NOT NULL.

```
mysql> select * from employee
    -> where salary is not null;
+--------+----------+---------+--------+--------------+--------+
| emp_id | emp_name | dept_id | salary | joining_date | city   |
+--------+----------+---------+--------+--------------+--------+
|    201 | Alice    |     101 |  71500 | 2024-04-03   | Mumbai |
|    202 | Bob      |     103 |  40000 | 2024-05-01   | Pune   |
|    203 | Casey    |     105 |  35000 | 2022-11-10   | Mumbai |
|    204 | Donald   |     102 |  60000 | 2022-08-20   | Dehli  |
|    205 | Emanual  |     101 |  77000 | 2023-02-15   | Dehli  |
|    206 | Franses  |     102 |  55000 | 2025-10-09   | Mumbai |
+--------+----------+---------+--------+--------------+--------+
6 rows in set (0.02 sec)
```

19. Write a query to calculate the total salary per department, but only for departments where total salary is greater than 1,00,000 (use HAVING).

```
mysql> select dept_id, sum(salary)
    -> from employee
    -> group by dept_id
    -> having sum(salary) > 100000;
+---------+-------------+
| dept_id | sum(salary) |
+---------+-------------+
|     101 |      148500 |
|     102 |      115000 |
+---------+-------------+
2 rows in set (0.00 sec)
```

20. Write a query to display all distinct employee names.

```
mysql> select distinct emp_name
    -> from employee;
+----------+
| emp_name |
+----------+
| Alice    |
| Bob      |
| Casey    |
| Donald   |
| Emanual  |
| Franses  |
+----------+
6 rows in set (0.02 sec)
```

21. Write a query to count the number of departments having the same location.

```
mysql> select location, count(dept_id)
    -> from department
    -> group by location;
+----------+----------------+
| location | count(dept_id) |
+----------+----------------+
| Mumbai   |              3 |
| Pune     |              1 |
| Dehli    |              1 |
+----------+----------------+
3 rows in set (0.02 sec)
```